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# OpenFOAM® Journal publication

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**Version(s)**: OpenFOAM® v20xx
**Repo**: https://github.com/xxx

Abstract. This is the place for an abstract.

## 1. Introduction

This is the place for introduction.

### 1.1. **Subsection.** Example text:

We shall consider the specific transport property $\phi$ and note that its spatial and temporal variation is governed by a second-order partical differential equation (PDE), viz.

$$\frac{\partial}{\partial t}(\rho\phi) + \nabla\bullet(\rho\phi\mathbf{U}) - \Gamma_\phi\nabla^2\phi - S_\phi(\phi) = 0. \tag{1}$$

Herein, $\phi = \phi(\mathbf{x}, t)$ is an arbitrary general intensive physical quantitity, e.g., a fluid property (scalar or tensor of any rank). Thus, Equation 1 is often referred to as generic transport equation.

OpenFOAM® (Open Field Operation And Manipulation) is a flexible and mature C++ Class Library for Computational Continuum Mechanics (CCM) and Multiphysics. Its Object-Oriented-Programming (OOP) paradigm enables to *mimic data types and basic operations* of CCM using top-level syntax as close as possible to the conventional mathematical notation *for tensors and partial differential equations*:

```
solve
(
  fvm::ddt(rho,Phi)
  + fvm::div(phi, Phi)
  - fvm::laplacian(Gamma, Phi)
 ==
  Sphi
);
```

Beside providing OpenFOAM® code itself, spatial and temporal discretisation of Equation 1 can be also described in a precise and concise manner using the finite-volume notation [1] - see Table 1.

## 2. Theoretical backgroud

Text in this section. Here is an examplary figure (Figure 1).

## 3. Conclusion

This is a conclusion.

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TABLE 1. Finite Volume Notation

| implicit differential operators | |
|---|---|
| rate of change | $\left[\!\left[\frac{\partial[\rho\phi]}{\partial t}\right]\!\right]$ |
| convection term | $\left[\!\left[\nabla\bullet\left(F[\phi]_{f(F,S,\gamma)}\right)\right]\!\right]$ |
| diffusion term | $[\![\nabla\bullet(\Gamma\nabla[\phi])]\!]$ |
| linear part of source term | $[\![S_p\,[\phi]]\!]$ |
| **explicit differential operators** | |
| temporal term | $\frac{\partial\rho\phi}{\partial t}$ |
| divergence term | $\nabla\bullet(\rho\mathbf{U}\phi_{f(\rho\mathbf{U},S,\gamma)})$ |
| laplacian term | $\nabla\bullet(\Gamma\nabla\phi)$ |
| constant part of source term | $S_u$ |

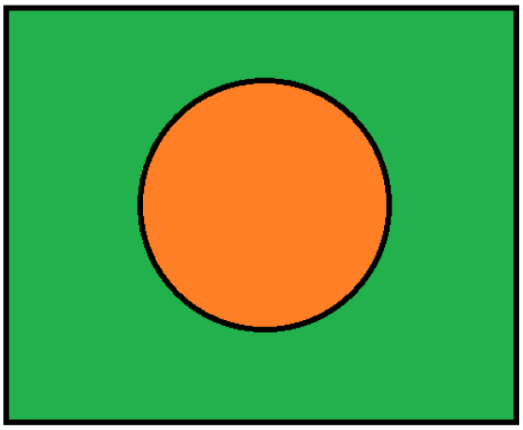


FIGURE 1. Examplary figure

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Please turn to http://img.mdpi.org/data/contributor-role-instruction.pdf (CRediT taxonomy) for the term explanation.

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